

# Lab07 – Deployment

## Rolling Update Deployment

In this exercise, you will create a Deployment with multiple replicas. After inspecting the Deployment, you will update its Pod template. Furthermore, you will use the rollout history to roll back to a previous revision.

1. Create a Deployment named `nginx` with 3 replicas. The Pods should use the `nginx:1.23.0` image and the name `nginx`. The Deployment uses the label `tier=backend`. The Pod template should use the label `app=v1`.

```
vagrant@kubernetes-control-plane:~/lab$ vim nginx-deployment.yaml
vagrant@kubernetes-control-plane:~/lab$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx
  labels:
    tier: backend
spec:
  replicas: 3
  selector:
    matchLabels:
      app: v1
  template:
    metadata:
      labels:
        app: v1
    spec:
      containers:
        - image: nginx:1.23.0
          name: nginx

vagrant@kubernetes-control-plane:~/lab$
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx created
vagrant@kubernetes-control-plane:~/lab$
```

2. List the Deployment and ensure that the correct number of replicas is running.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl get deployment,pod
NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/nginx               3/3      3              3            2m31s

NAME                                READY    STATUS    RESTARTS    AGE
pod/nginx-5d44b9f45b-dhlnr          1/1      Running   0            2m31s
pod/nginx-5d44b9f45b-mzfjp          1/1      Running   0            2m31s
pod/nginx-5d44b9f45b-snmvk          1/1      Running   0            2m31s
vagrant@kubernetes-control-plane:~/lab$
```

3. Update the image to `nginx:1.23.4`.

```
vagrant@k8s-control-plane:~/lab$ kubectl set image deployment/nginx nginx=nginx:1.23.4
deployment.apps/nginx image updated
vagrant@k8s-control-plane:~/lab$ kubectl get deployment,pod
NAME READY UP-TO-DATE AVAILABLE AGE
deployment.apps/nginx 3/3 1 3 3m47s

NAME READY STATUS RESTARTS AGE
pod/nginx-5484b6d87b-fl2g5 0/1 ContainerCreating 0 18s
pod/nginx-5d44b9f45b-dhlnr 1/1 Running 0 3m47s
pod/nginx-5d44b9f45b-mzfjp 1/1 Running 0 3m47s
pod/nginx-5d44b9f45b-snmvk 1/1 Running 0 3m47s
vagrant@k8s-control-plane:~/lab$
```

```
vagrant@k8s-control-plane:~/lab$ kubectl get deployment,pod
NAME READY UP-TO-DATE AVAILABLE AGE
deployment.apps/nginx 3/3 2 3 4m56s

NAME READY STATUS RESTARTS AGE
pod/nginx-5484b6d87b-fl2g5 1/1 Running 0 87s
pod/nginx-5484b6d87b-h749n 0/1 ContainerCreating 0 37s
pod/nginx-5d44b9f45b-mzfjp 1/1 Running 0 4m56s
pod/nginx-5d44b9f45b-snmvk 1/1 Running 0 4m56s
vagrant@k8s-control-plane:~/lab$
```

#### 4. Verify that the change has been rolled out to all replicas.

```
vagrant@k8s-control-plane:~/lab$ kubectl get deployment,pod
NAME READY UP-TO-DATE AVAILABLE AGE
deployment.apps/nginx 3/3 3 3 5m59s

NAME READY STATUS RESTARTS AGE
pod/nginx-5484b6d87b-6tgfr 1/1 Running 0 42s
pod/nginx-5484b6d87b-fl2g5 1/1 Running 0 2m30s
pod/nginx-5484b6d87b-h749n 1/1 Running 0 100s
vagrant@k8s-control-plane:~/lab$ kubectl rollout history deployment nginx
deployment.apps/nginx
REVISION CHANGE-CAUSE
1 <none>
2 <none>

vagrant@k8s-control-plane:~/lab$ kubectl rollout history deployment nginx --revision=2
deployment.apps/nginx with revision #2
Pod Template:
  Labels: app=v1
          pod-template-hash=5484b6d87b
  Containers:
    nginx:
      Image: nginx:1.23.4
      Port: <none>
      Host Port: <none>
      Environment: <none>
      Mounts: <none>
      Volumes: <none>
vagrant@k8s-control-plane:~/lab$
```

#### 5. Assign the change cause "Pick up patch version" to the revision.

```
vagrant@k8s-control-plane:~/lab$ kubectl annotate deployment nginx kubernetes.io/change-cause="Pick up patch version"
deployment.apps/nginx annotated
vagrant@k8s-control-plane:~/lab$ kubectl rollout history deployment nginx
deployment.apps/nginx
REVISION CHANGE-CAUSE
1 <none>
2 Pick up patch version
vagrant@k8s-control-plane:~/lab$
vagrant@k8s-control-plane:~/lab$
```

## 6. Scale the Deployment to 5 replicas.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl scale deployment nginx --replicas=5
deployment.apps/nginx scaled
vagrant@kubernetes-control-plane:~/lab$
vagrant@kubernetes-control-plane:~/lab$ kubectl get pod -l app=v1
NAME                                READY   STATUS    RESTARTS   AGE
nginx-5484b6d87b-6tgfr              1/1     Running   0           4m53s
nginx-5484b6d87b-fl2g5              1/1     Running   0           6m41s
nginx-5484b6d87b-h749n              1/1     Running   0           5m51s
nginx-5484b6d87b-s4fxk              1/1     Running   0           24s
nginx-5484b6d87b-xf8gv              1/1     Running   0           24s
vagrant@kubernetes-control-plane:~/lab$
vagrant@kubernetes-control-plane:~/lab$
```

## 7. Have a look at the Deployment rollout history.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl rollout history deployment nginx
deployment.apps/nginx
REVISION  CHANGE-CAUSE
1         <none>
2         Pick up patch version
vagrant@kubernetes-control-plane:~/lab$
```

## 8. Revert the Deployment to revision 1.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl rollout undo deployment/nginx --to-revision=1
deployment.apps/nginx rolled back
vagrant@kubernetes-control-plane:~/lab$ kubectl rollout history deployment nginx
deployment.apps/nginx
REVISION  CHANGE-CAUSE
2         Pick up patch version
3         <none>
vagrant@kubernetes-control-plane:~/lab$
vagrant@kubernetes-control-plane:~/lab$
```

## 9. Ensure that the Pods use the image `nginx:1.23.0`.

```
vagrant@kubernetes-control-plane:~/lab$ kubectl rollout history deployment nginx --revision=3
deployment.apps/nginx with revision #3
Pod Template:
  Labels:      app=v1
              pod-template-hash=5d44b9f45b
  Containers:
    nginx:
      Image:      nginx:1.23.0
      Port:      <none>
      Host Port:  <none>
      Environment: <none>
      Mounts:     <none>
      Volumes:    <none>
vagrant@kubernetes-control-plane:~/lab$
```

## Blue/Green Deployment

In this exercise, you will set up a blue-green Deployment scenario. You'll first create the initial (blue) Deployment and expose it with a Service. Later, you will create a second (green) Deployment and switch over traffic.

10. Create a Deployment named `nginx-blue` with 3 replicas. The Pod template of the Deployment should use container image `nginx:1.23.0` and assign the label `version=blue`.

```
vagrant@kubernetes-control-plane:~/lab$ vim blue-deployment.yaml
vagrant@kubernetes-control-plane:~/lab$ cat blue-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-blue
spec:
  replicas: 3
  selector:
    matchLabels:
      version: blue
  template:
    metadata:
      labels:
        version: blue
    spec:
      containers:
      - image: nginx:1.23.0
        name: nginx
        ports:
        - containerPort: 80
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f blue-deployment.yaml
deployment.apps/nginx-blue created
vagrant@kubernetes-control-plane:~/lab$ kubectl get pods -l version=blue
NAME                                READY   STATUS    RESTARTS   AGE
nginx-blue-6787c47cc9-dbhvb         1/1     Running   0           38s
nginx-blue-6787c47cc9-g7zhh         1/1     Running   0           38s
nginx-blue-6787c47cc9-qmz8f         1/1     Running   0           38s
vagrant@kubernetes-control-plane:~/lab$
```

11. Expose the Deployment with a Service of type `ClusterIP` named `nginx`. Map the incoming and outgoing port to 80. Select the Pod with label `version=blue`.

```
vagrant@kubernetes-control-plane:~/lab$ vim service.yaml
vagrant@kubernetes-control-plane:~/lab$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx
spec:
  selector:
    version: blue
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
vagrant@kubernetes-control-plane:~/lab$ kubectl apply -f service.yaml
service/nginx created
vagrant@kubernetes-control-plane:~/lab$ kubectl get deploy,svc,pod
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/nginx-blue          3/3     3             3           2m8s

NAME                                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
service/kubernetes                  ClusterIP   10.96.0.1    <none>        443/TCP    4m2s
service/nginx                       ClusterIP   10.105.34.206 <none>        80/TCP     33s

NAME                                READY   STATUS    RESTARTS   AGE
pod/nginx-blue-6787c47cc9-dbhvb     1/1     Running   0           2m8s
pod/nginx-blue-6787c47cc9-g7zhh     1/1     Running   0           2m8s
pod/nginx-blue-6787c47cc9-qmz8f     1/1     Running   0           2m8s
vagrant@kubernetes-control-plane:~/lab$
```

12. Run a temporary Pod with the container image `alpine/curl:3.14` to make a call against the Service using `curl`.

```
vagrant@k8s-control-plane:~/lab$ kubectl run tmp --image=alpine/curl:3.14 --restart=Never -it --rm -- curl -sI nginx.default.svc.cluster.local | grep Server
Server: nginx/1.23.0
```

13. Create a second Deployment named `nginx-green` with 3 replicas. The Pod template of the Deployment should use container image `nginx:1.23.4` and assign the label `version=green`.

```
vagrant@k8s-control-plane:~/lab$ vim green-deployment.yaml
vagrant@k8s-control-plane:~/lab$ cat green-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-green
spec:
  replicas: 3
  selector:
    matchLabels:
      version: green
  template:
    metadata:
      labels:
        version: green
    spec:
      containers:
      - image: nginx:1.23.4
        name: nginx
        ports:
        - containerPort: 80
vagrant@k8s-control-plane:~/lab$ kubectl apply -f green-deployment.yaml
deployment.apps/nginx-green created
vagrant@k8s-control-plane:~/lab$ kubectl get pods -l version=green
NAME                                READY   STATUS    RESTARTS   AGE
nginx-green-88c6d7c9d-jskz7         1/1     Running   0           16s
nginx-green-88c6d7c9d-k7fxs         1/1     Running   0           16s
nginx-green-88c6d7c9d-r442h         1/1     Running   0           16s
vagrant@k8s-control-plane:~/lab$
```

14. Change the Service's label selection so that traffic will be routed to the Pods controlled by the Deployment `nginx-green`.

```
vagrant@k8s-control-plane:~/lab$ vim service.yaml
vagrant@k8s-control-plane:~/lab$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx
spec:
  selector:
    version: green
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
vagrant@k8s-control-plane:~/lab$ kubectl apply -f service.yaml
service/nginx configured
vagrant@k8s-control-plane:~/lab$
```

## 15. Delete the Deployment named `nginx-blue`.

```
vagrant@k8s-control-plane:~/lab$ kubectl delete deployment nginx-blue
deployment.apps "nginx-blue" deleted
vagrant@k8s-control-plane:~/lab$ kubectl get all -o wide
```

| NAME                            | READY | STATUS  | RESTARTS | AGE  | IP        | NODE       | NOMINATED NODE | READINESS GATES |
|---------------------------------|-------|---------|----------|------|-----------|------------|----------------|-----------------|
| pod/nginx-green-88c6d7c9d-jskz7 | 1/1   | Running | 0        | 3m6s | 10.44.0.2 | kube-node1 | <none>         | <none>          |
| pod/nginx-green-88c6d7c9d-k7fxs | 1/1   | Running | 0        | 3m6s | 10.36.0.3 | kube-node2 | <none>         | <none>          |
| pod/nginx-green-88c6d7c9d-r442h | 1/1   | Running | 0        | 3m6s | 10.44.0.3 | kube-node1 | <none>         | <none>          |

  

| NAME               | TYPE      | CLUSTER-IP    | EXTERNAL-IP | PORT(S) | AGE  | SELECTOR      |
|--------------------|-----------|---------------|-------------|---------|------|---------------|
| service/kubernetes | ClusterIP | 10.96.0.1     | <none>      | 443/TCP | 10m  | <none>        |
| service/nginx      | ClusterIP | 10.105.34.206 | <none>      | 80/TCP  | 7m7s | version=green |

  

| NAME                        | READY | UP-TO-DATE | AVAILABLE | AGE  | CONTAINERS | IMAGES       | SELECTOR      |
|-----------------------------|-------|------------|-----------|------|------------|--------------|---------------|
| deployment.apps/nginx-green | 3/3   | 3          | 3         | 3m6s | nginx      | nginx:1.23.4 | version=green |

  

| NAME                                  | DESIRED | CURRENT | READY | AGE  | CONTAINERS | IMAGES       | SELECTOR                                  |
|---------------------------------------|---------|---------|-------|------|------------|--------------|-------------------------------------------|
| replicaset.apps/nginx-green-88c6d7c9d | 3       | 3       | 3     | 3m6s | nginx      | nginx:1.23.4 | pod-template-hash=88c6d7c9d,version=green |

```
vagrant@k8s-control-plane:~/lab$
```

## 16. Run a temporary Pod with the container image `alpine/curl:3.14` to make a call against the Service.

```
vagrant@k8s-control-plane:~/lab$ kubectl run tmp --image=alpine/curl:3.14 --restart=Never -it --rm -- curl -sI nginx.default.svc.cluster.local | grep Server
Server: nginx/1.23.4
```